

Chapter 6

Unemployment: Search and Efficiency Wages

Learning Objectives, Part I

- 6.1** List the key labor market facts concerning the unemployment rate, the participation rate, and the employment/population ratio.
- 6.2** Describe the Beveridge curve, and explain its importance.
- 6.3** In the one-sided search model, explain how the reservation wage is determined.

Learning Objectives, Part II

- 6.4** Show how the one-sided search model determines the unemployment rate.
- 6.5** Use the one-sided search model to determine the effects of changes in the labor market on the efficiency wage and the unemployment rate.

Learning Objectives, Part III

- 6.6** Construct an equilibrium in the two-sided search model.
- 6.7** Use the two-sided search model to explain how shocks to the labor market change labor force participation, unemployment, vacancies, aggregate output, and labor market tightness.

Key Labor Market Variables

N = working age population

Q = labor force (employed plus unemployed)

U = unemployed

Unemployment rate = $\frac{U}{Q}$

Participation rate = $\frac{Q}{N}$

Employment/population ratio = $\frac{Q-U}{N}$

Figure 6.1 The Unemployment Rate

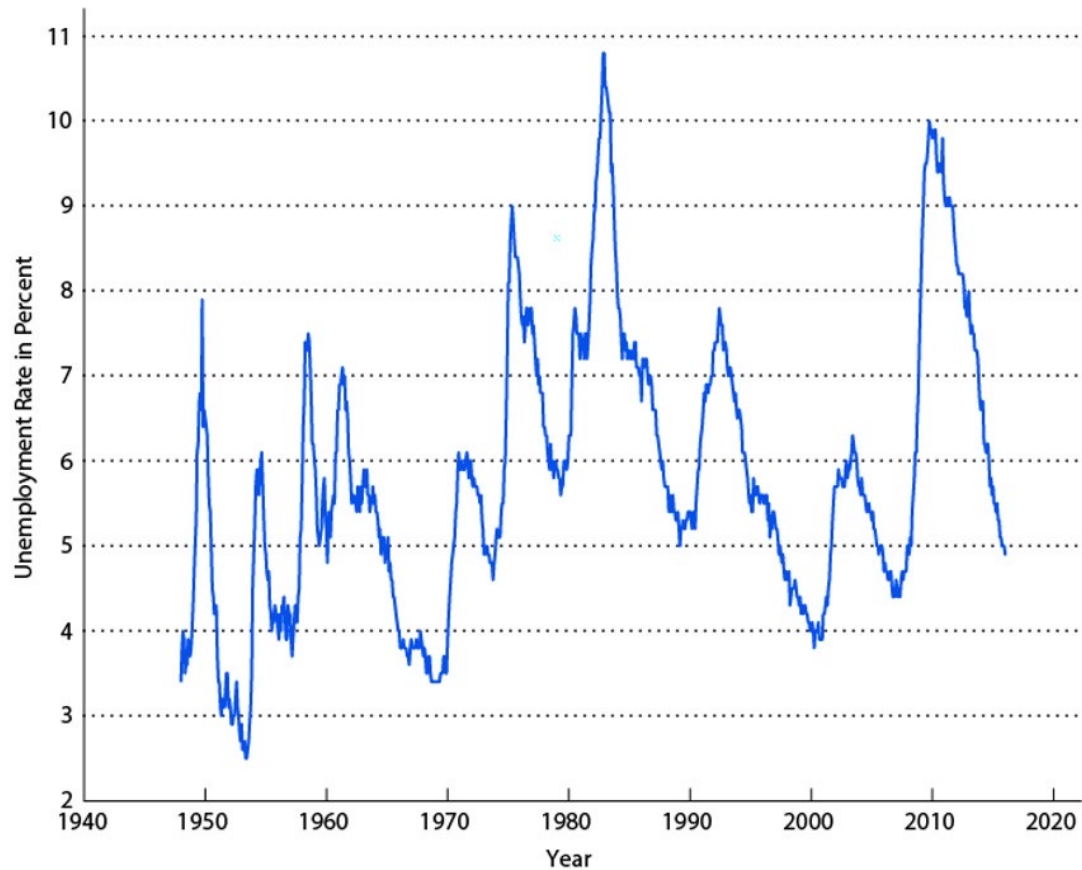


Figure 6.2 Deviations From Trend in the Unemployment Rate and Real GDP

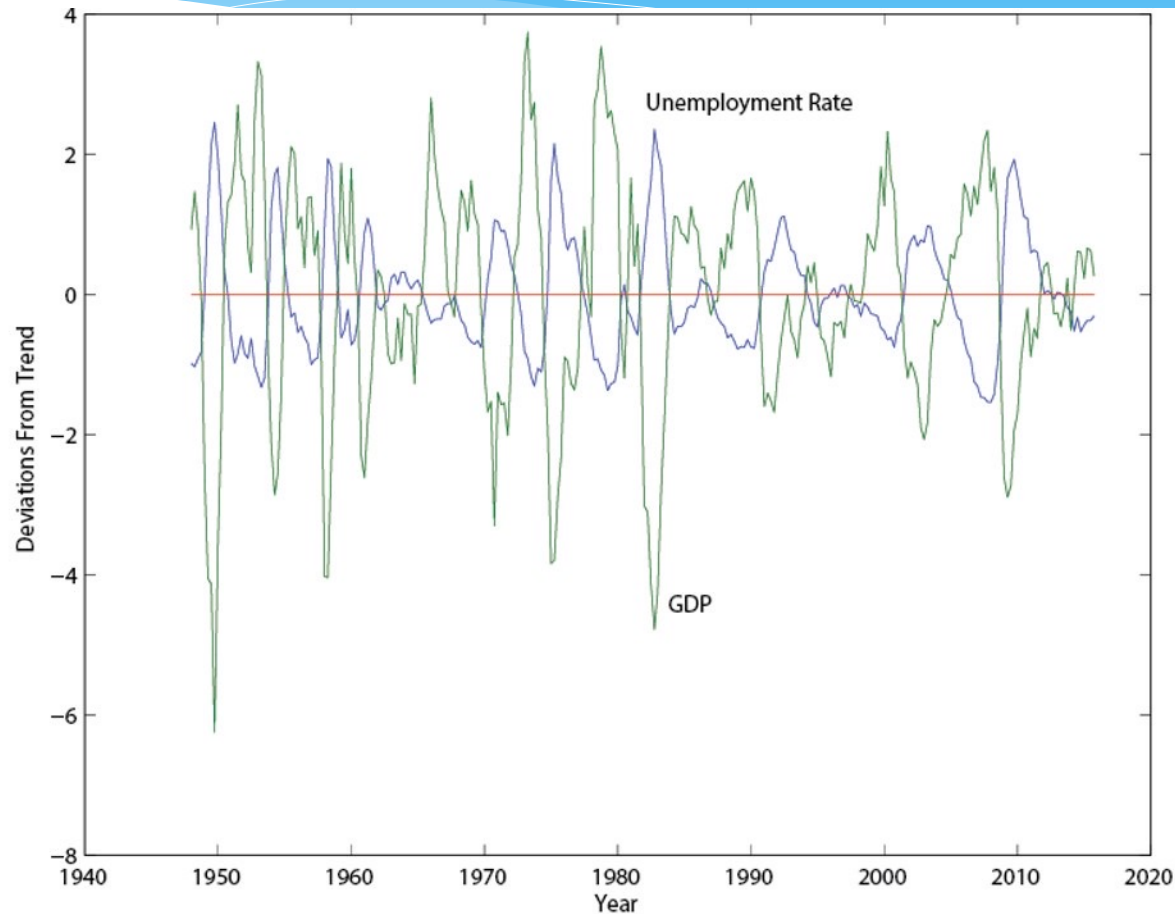


Figure 6.3 Labor Force Participation Rate

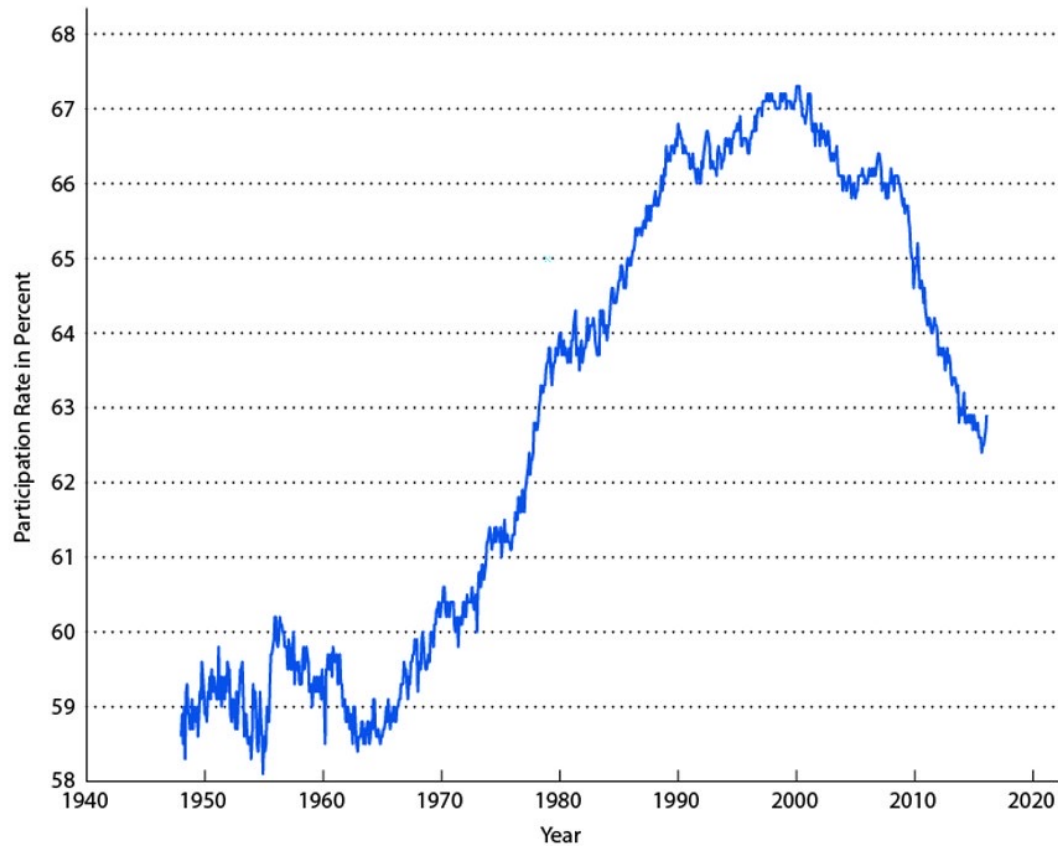


Figure 6.4 Labor Force Participation Rates of Men and Women

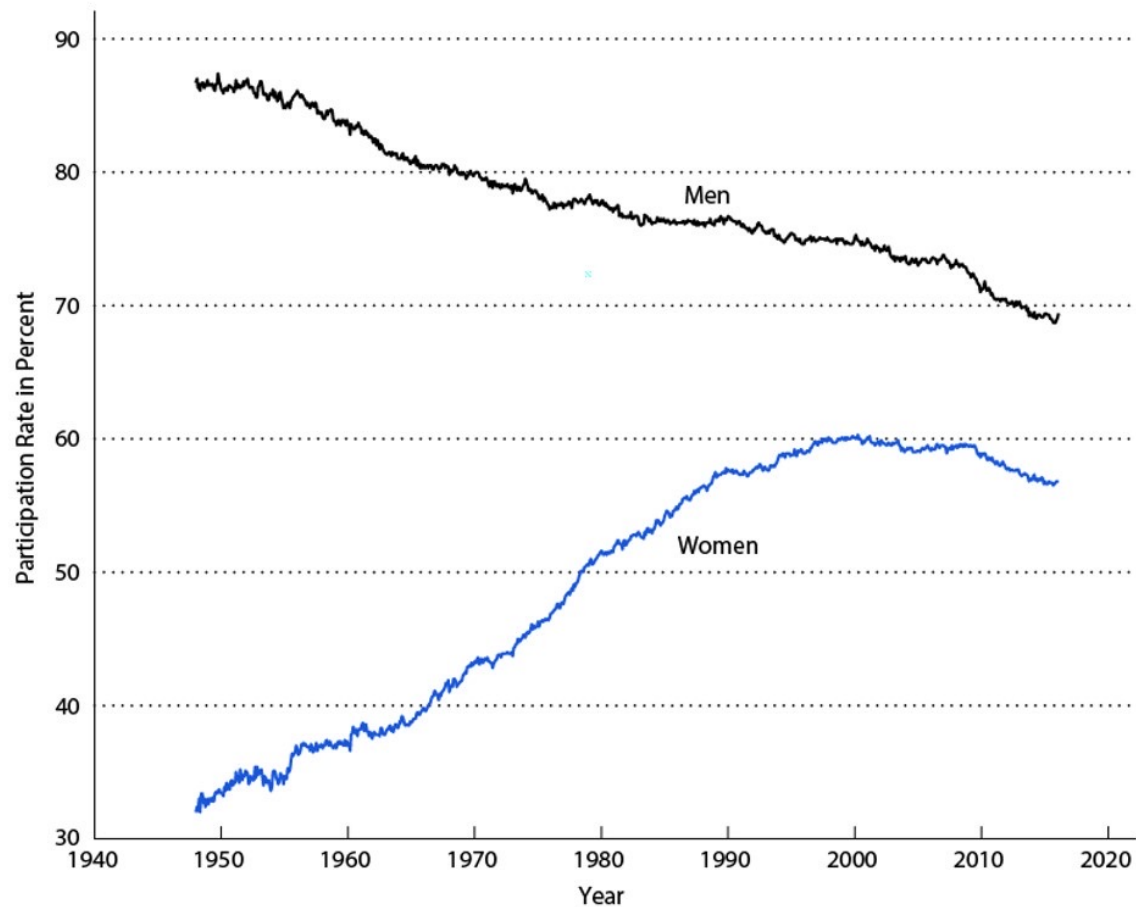


Figure 6.5 Percentage Deviations From Trend: Labor Force Participation Rate and Real GDP

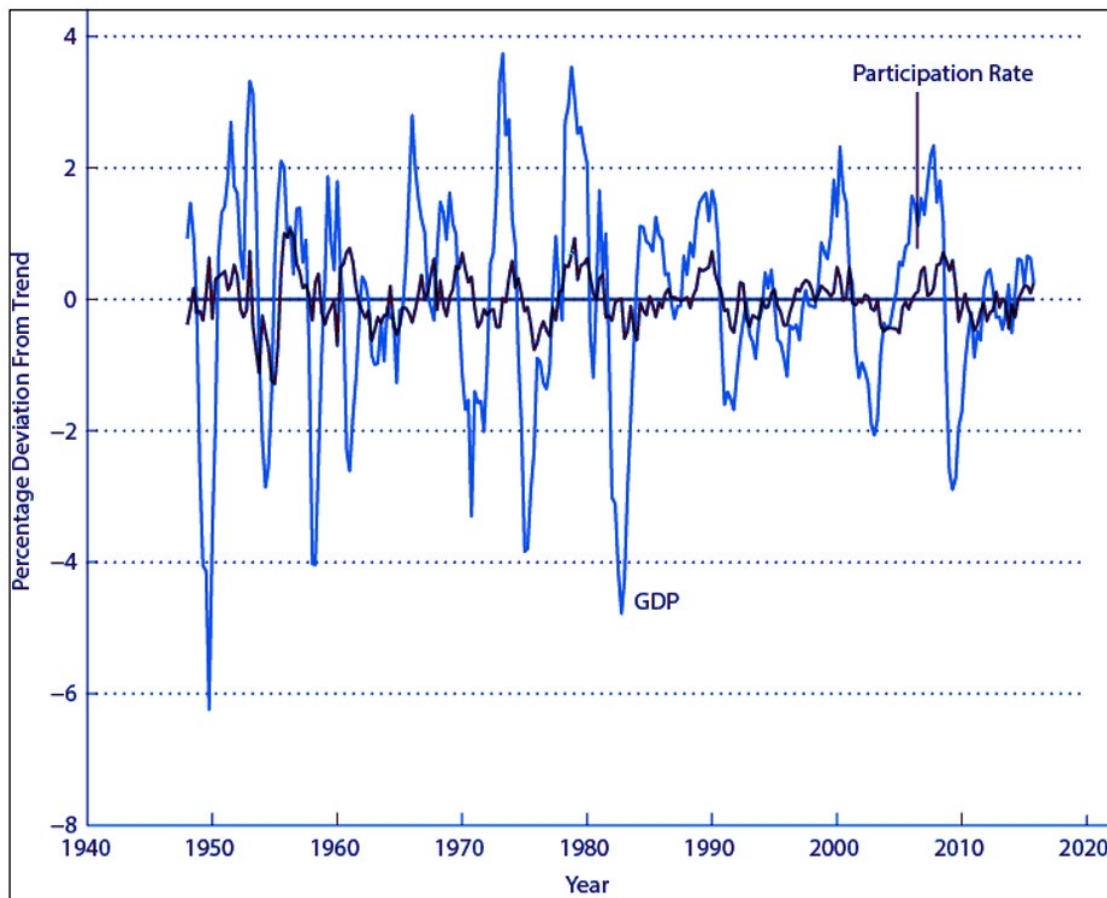
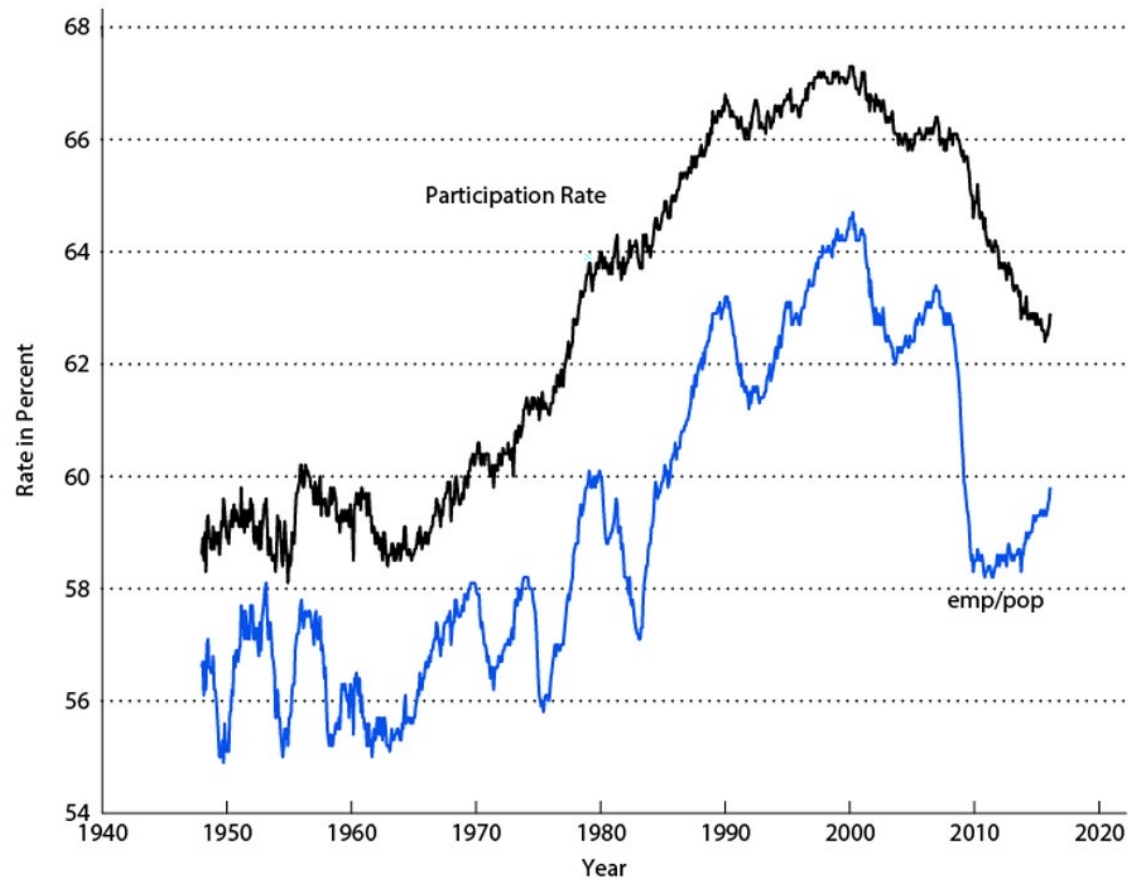


Figure 6.6 Labor Force Participation Rate and Employment/Population Ratio



Vacancies and Unemployment

- A = aggregate number of vacancies listed by firms.
- Vacancy rate = $\frac{A}{A+Q-U}$

Figure 6.7 The Vacancy Rate and Unemployment Rate

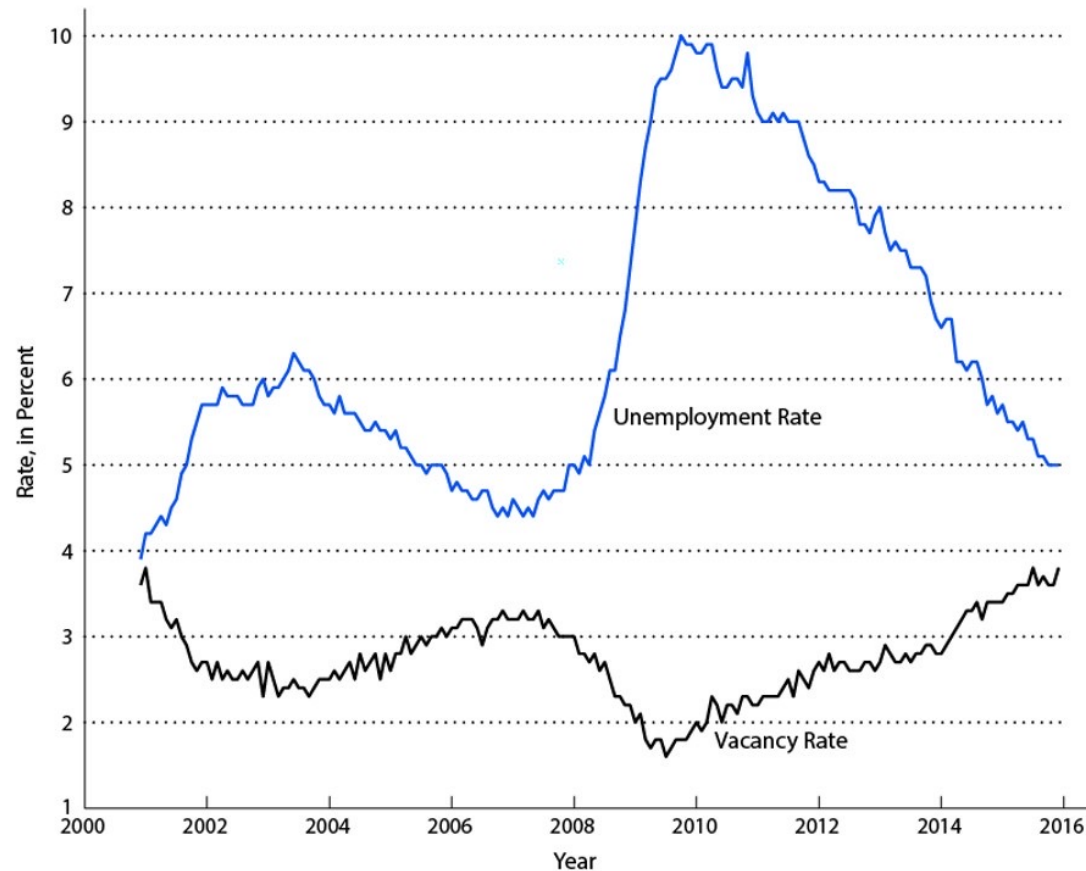
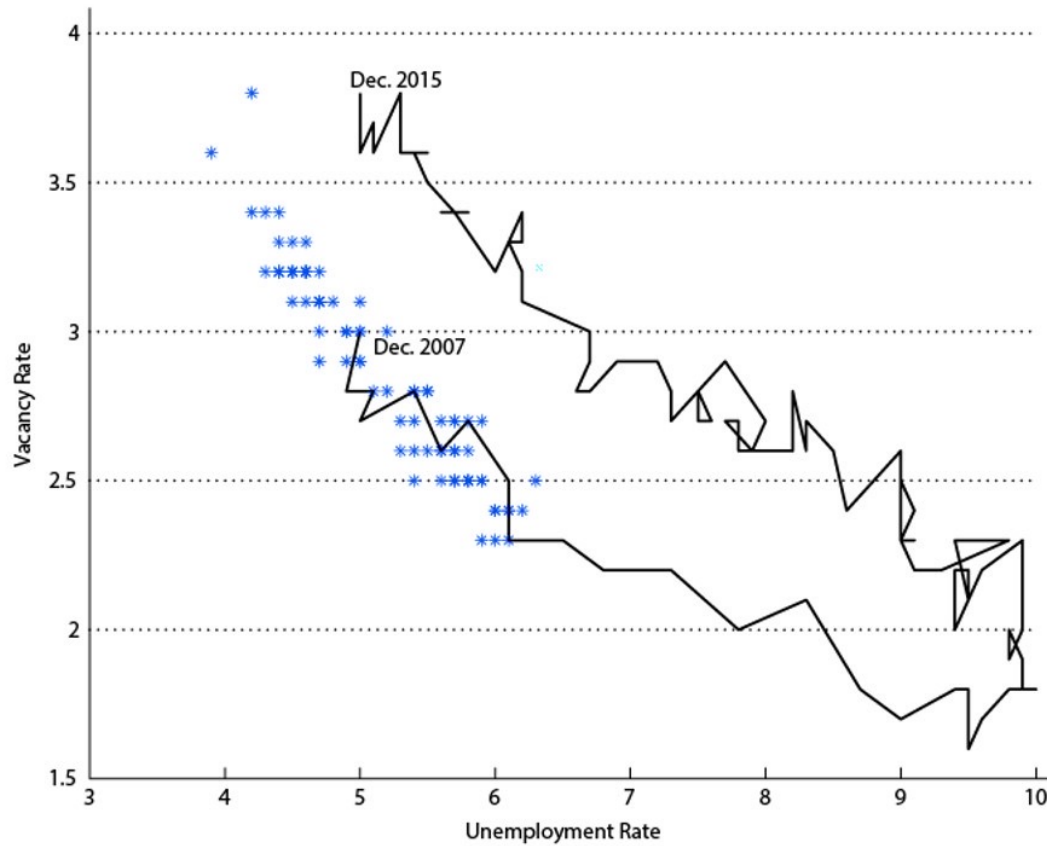


Figure 6.8 Beveridge Curve



Key Labor Market Observations

- The unemployment rate is countercyclical.
- The unemployment rate and the vacancy rate are negatively correlated (the Beveridge curve).
- The Beveridge curve shifted out during the last recession.

The One-Sided Search Model

- Focuses on the behavior of an unemployed worker.
- Unemployed worker receives wage offer with probability p .
- Wage offer w either accepted, or unemployed worker turns down the offer and continues to search.
- All workers are either employed or unemployed. Not-in-the-labor-force not modeled.

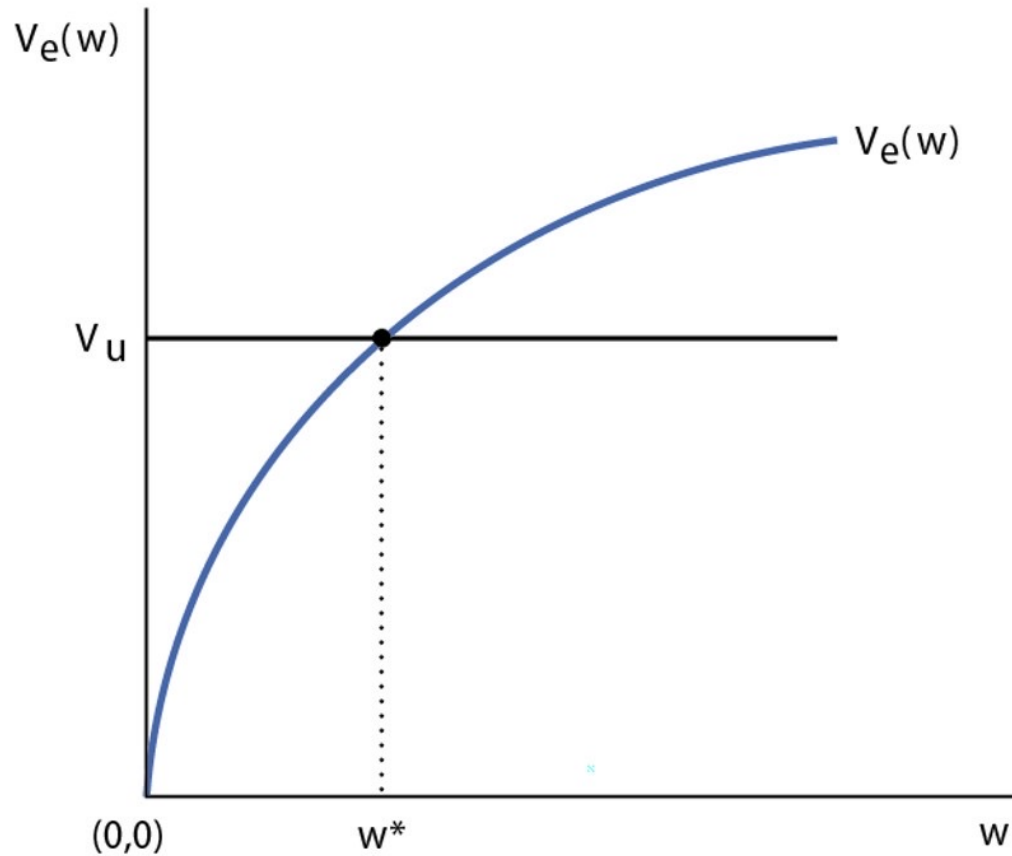
Welfare of the Employed and Unemployed

- * Welfare of an employed worker $V_e(w)$:
 - * decreases with the separation rate s .
- Welfare of an unemployed worker V_u :
 - increases with the UI benefit, b .
 - increases with p , the frequency with which the unemployed worker receives job offers.

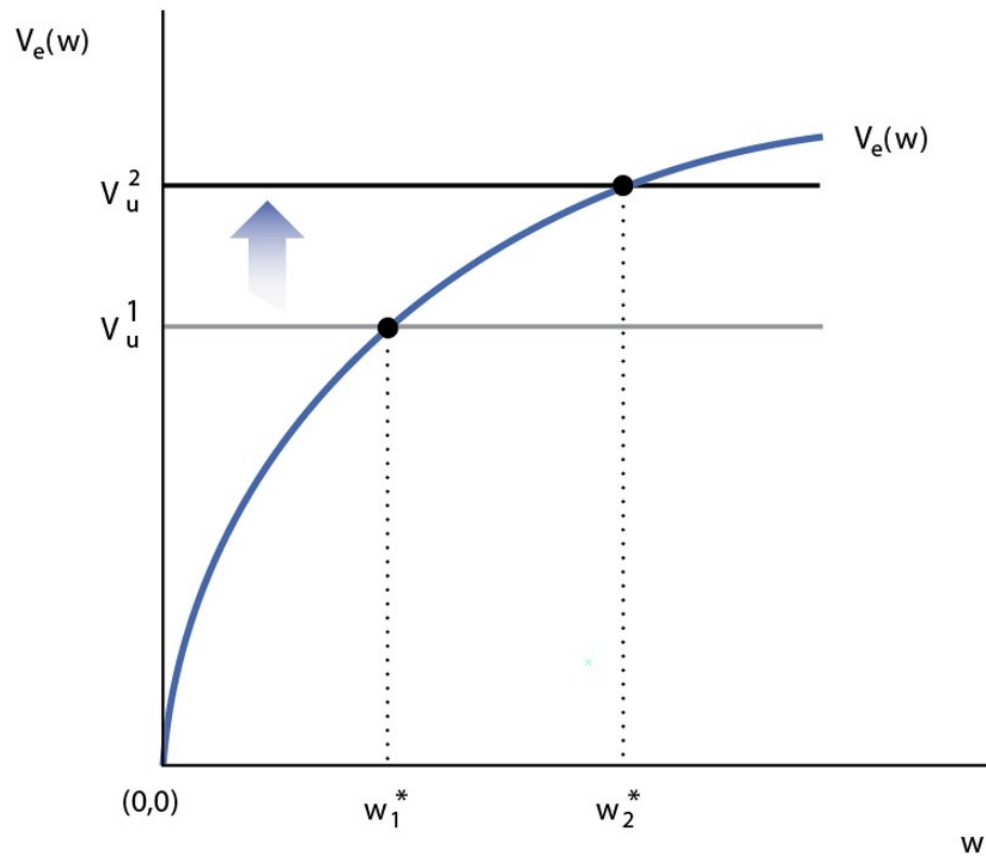
Reservation Wage (1 of 2)

- * Reservation Wage: the wage w^* at which the unemployed worker is just indifferent between accepting and declining a job offer.
- The unemployed worker turns down jobs with $w < w^*$, and accepts jobs with $w \geq w^*$.

Reservation Wage



An Increase in the Unemployment Insurance Benefit

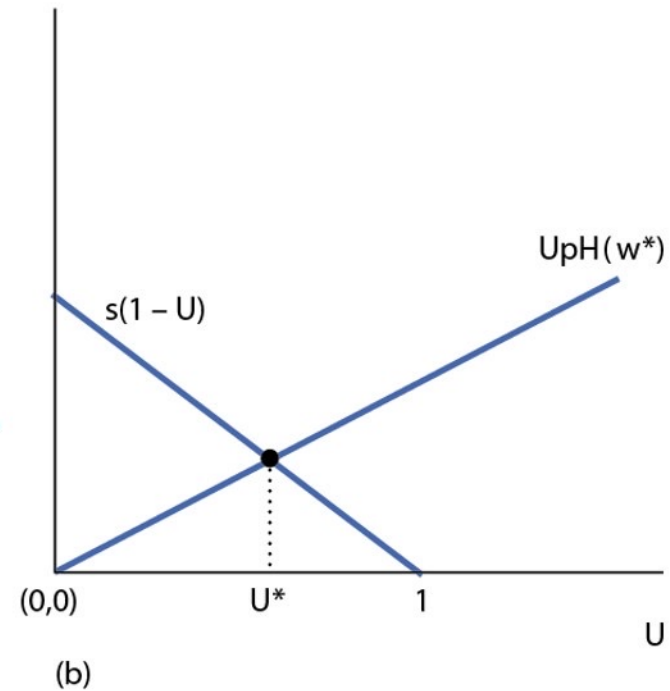
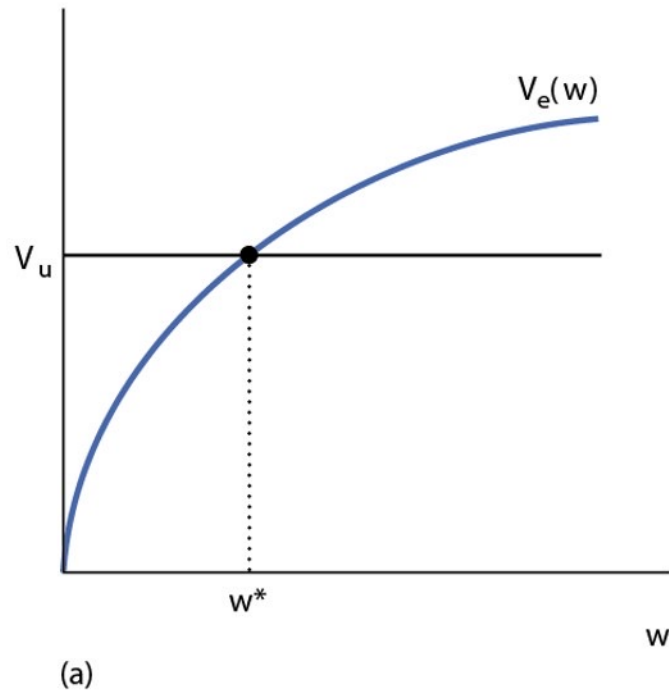


Determining the Unemployment Rate in the One-Sided Search Model

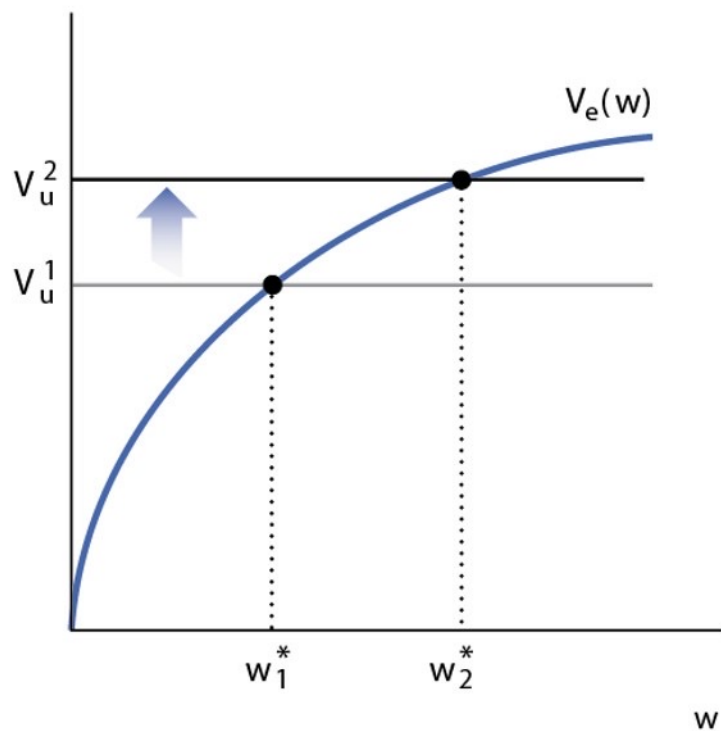
- * $H(w)$ = Fraction of workers receiving a wage offer greater than w .
- * U = unemployment rate.
- * Long-run equilibrium: flow of workers from employment to unemployment equals the flow in reverse direction:

$$s(1 - U) = UpH(w^*)$$

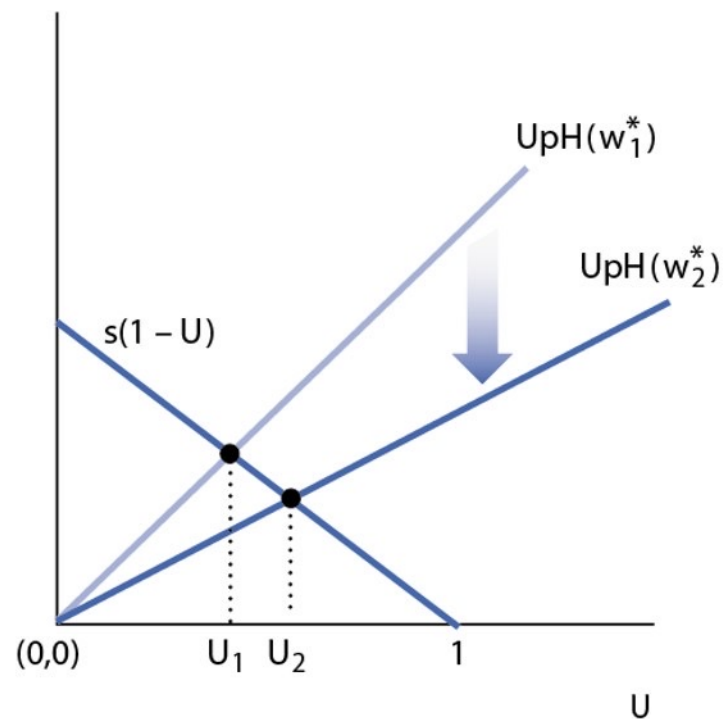
Determining the Reservation Wage and Unemployment Rate



An Increase in the UI Benefit

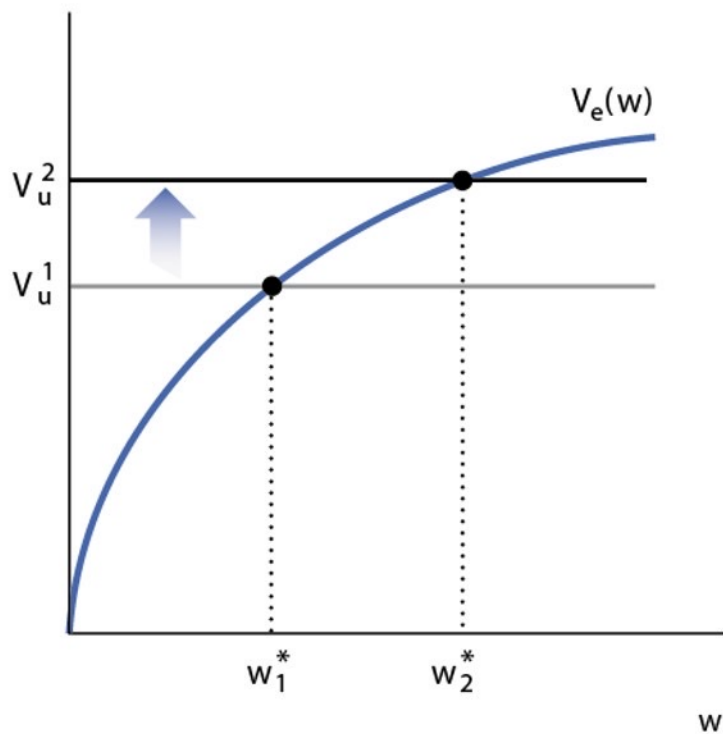


(a)

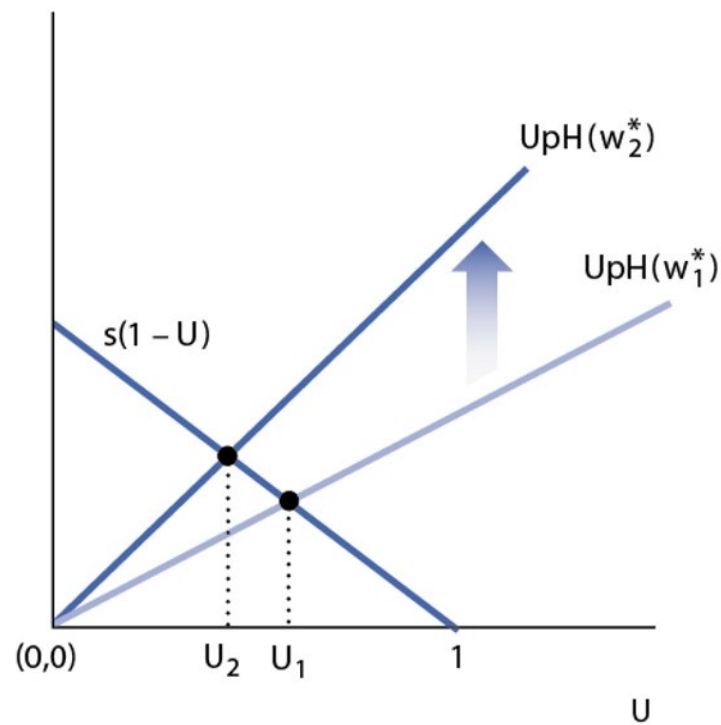


(b)

An Increase in the Job Offer Rate



(a)



(b)

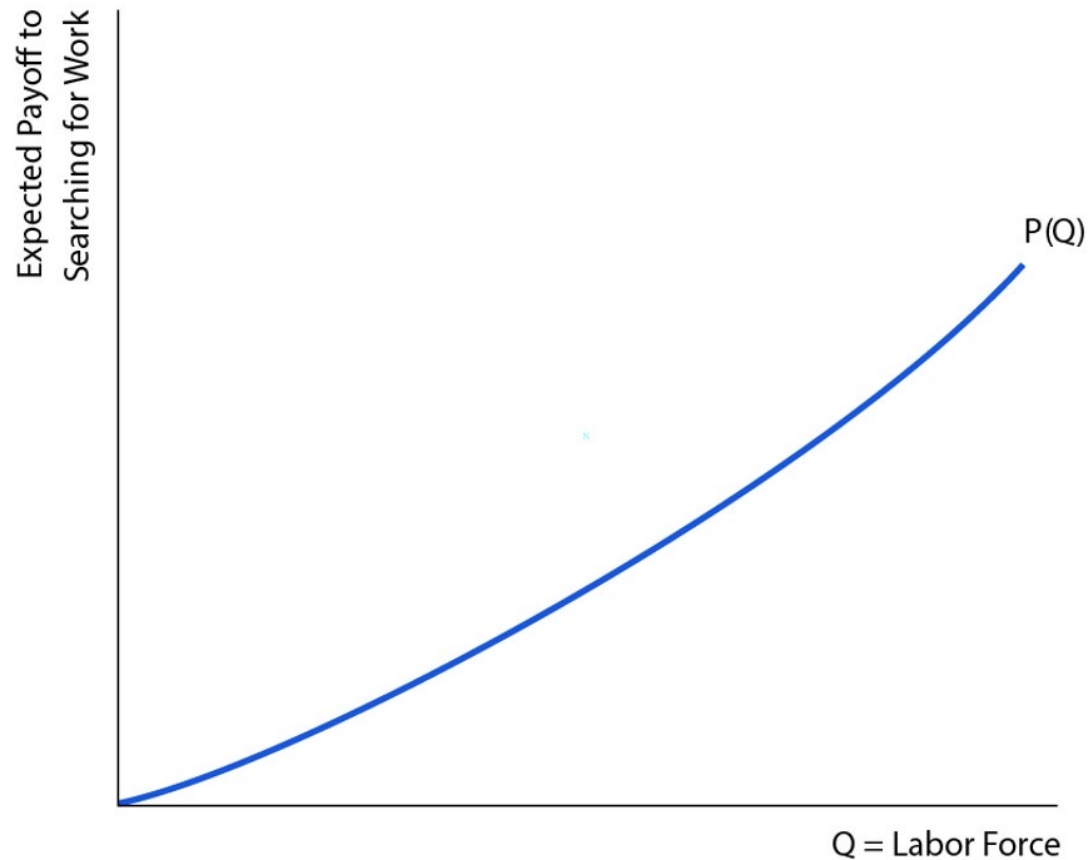
A Two-Sided Search Model of Unemployment

- One-period model.
- N consumers who can all potentially work, so N is the working age population.
- Number of firms is endogenous.

Consumers in the Two-Sided Search Model

- Each of the N consumers chooses whether to work outside the market (homework), or to search for work in the market.
- Q = number of consumers who search for work
- $N-Q$ = not in the labor force
- $P(Q)$ = expected payoff to searching for work that would induce Q workers to search. $P(Q)$ is essentially the supply curve for searching workers.

Figure 6.17 The Supply Curve of Consumers Searching for Work



Firms

- A firm must post a vacancy in order to have a chance of matching with a worker.
- k = cost of posting a vacancy, in units of consumption goods
- A = number of active firms (firms posting vacancies)

Matching

- A successful match in the model is between one worker and one firm.
- M = aggregate number of matches
- e = matching efficiency
- Matching function:

$$M = em(Q, A)$$

Properties of the Matching Function

- The matching function has properties like a production function.
- The “inputs,” Q and A , produce the “output” M , and e plays the same role as total factor productivity in the production function.
- The matching function has constant returns to scale, positive marginal products, and diminishing marginal products.

Supply Side of the Labor Market: Optimization by Consumers

- * Each consumer chooses between home production and searching for work.
- If the consumer chooses to search for work, then he or she finds a match with a firm with probability

$$p_c = \frac{em(Q, A)}{Q}$$

- If the consumer searches for work and is matched he/she receives wage w .
- If the consumer searches and is not matched, then he/she is unemployed and receives the UI benefit b .

Marginal Consumer

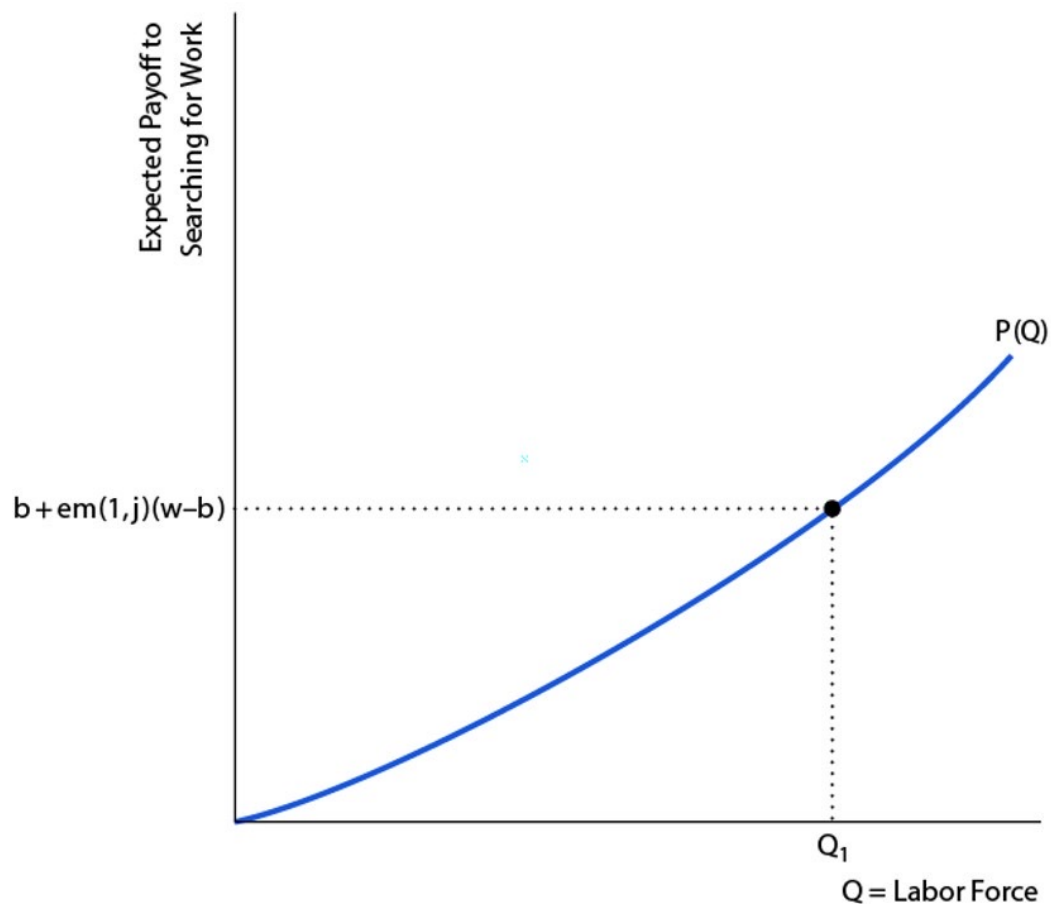
- For the consumer who is indifferent between home production and searching for work,

$$P(Q) = b + em(1, j)(w - b)$$

- Here, j is *labor market tightness*,

$$j = \frac{A}{\dots}$$

Figure 6.18 The Supply Side of the Labor Market



Demand Side of the Labor Market

- A firm entering the labor market bears the cost k to post a vacancy.
- The probability that a firm with a vacancy finds a worker to fill the job is

$$p_f = em \left(\frac{1}{j}, 1 \right)$$

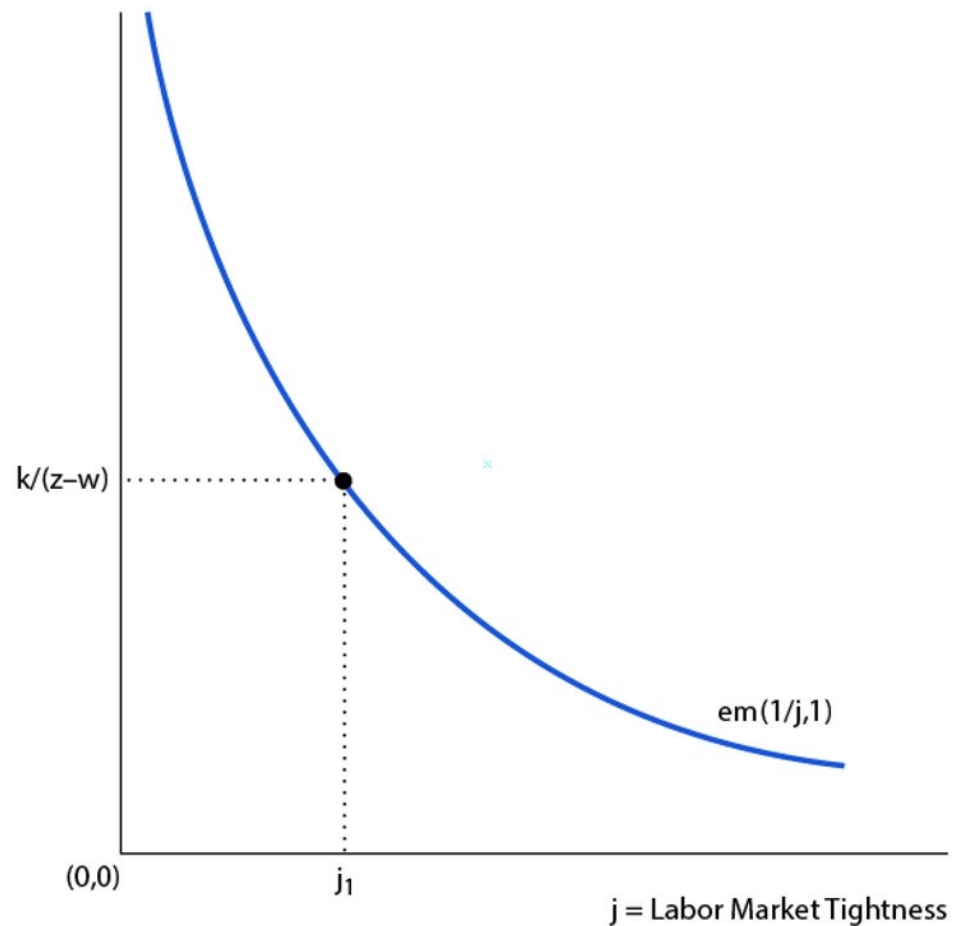
- When matched, a worker and firm produce z , so the payoff to the firm is $\text{profit} = z - w$.

Expected Net Payoff for a Firm Posting a Vacancy is Zero in Equilibrium

- In equilibrium, k must be equal to the expected payoff for the firm from posting the vacancy, which implies:

$$em\left(\frac{1}{j}, 1\right) = \frac{k}{z - w}$$

Figure 6.19 Demand Side of the Labor Market



Nash Bargaining

- Use Nash bargaining theory to determine how a matched firm and worker split the total revenue from production.
- Worker's surplus = $w - b$ (wage minus UI benefit)
- Firm's surplus = $z - w$ (profit)
- Total surplus = $z - b$
- a = worker's share of total surplus (“bargaining power”)

$$w = az + (1 - a)b$$

Equilibrium

- Two equations determining Q and j (from supply side, demand side, and Nash bargaining):

$$P(Q) = b + em(1, j)a(z - b)$$

$$em\left(\frac{1}{1}\right) = \frac{k}{\dots}$$

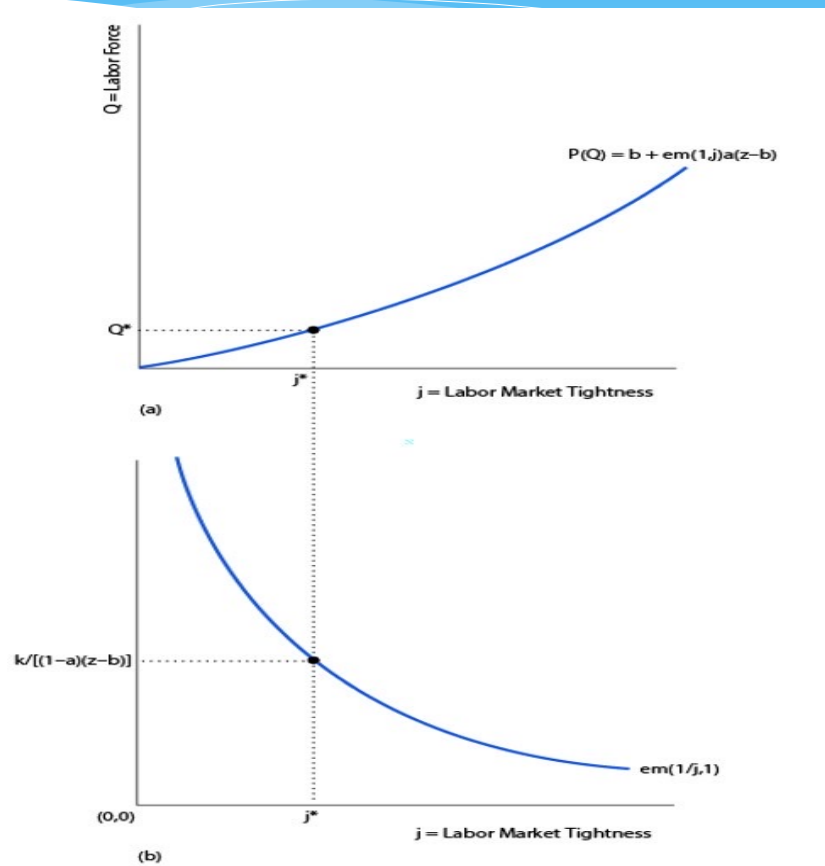
Equilibrium Unemployment Rate, Vacancy Rate, and Aggregate Output

- In equilibrium, as functions of j and Q , the unemployment rate, vacancy rate, and level of aggregate output, respectively, are:

$$u = 1 - em(1, j)$$

$$v = 1 - em\left(\frac{1}{j}, 1\right)$$

Figure 6.20 Equilibrium in the Two-Side Search Model



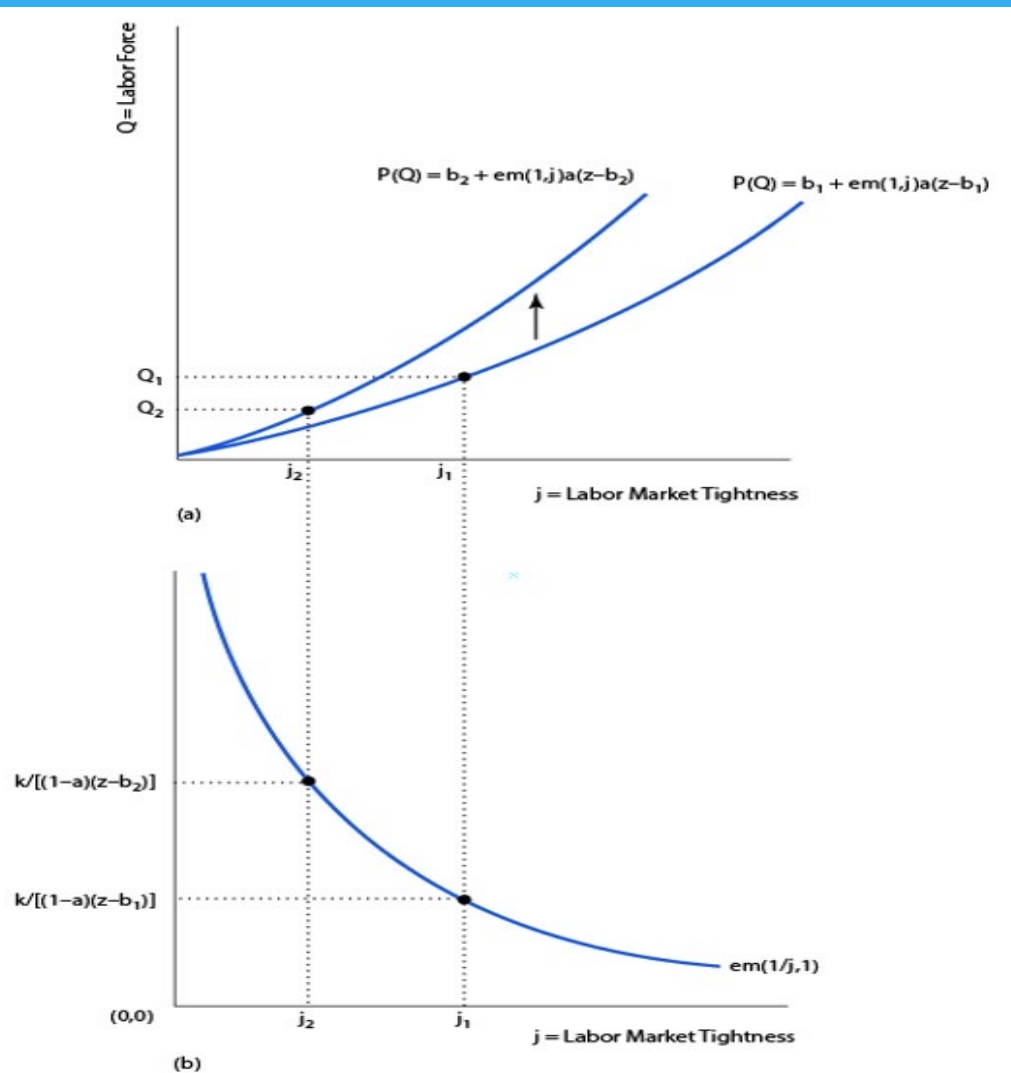
Working with the Two-Sided Search Model: 3 Experiments

- * Increase in the UI benefit b .
- * Increase in productivity z .
- * Decrease in matching efficiency e .

Increase in the UI Benefit, b

- Reduces total surplus from a match, $z - b$.
- Increases the wage, w , as the alternative to working becomes more tempting for a searching consumer.
- Posting vacancies becomes less attractive for firms, so labor market tightness, j , falls.
- For consumers, searching for work becomes more attractive, as the wage is higher. But searching for work is also less attractive, as the chances of finding a job are lower (j is lower).
- Q may rise or fall given these two opposing effects.
- u rises and v falls.

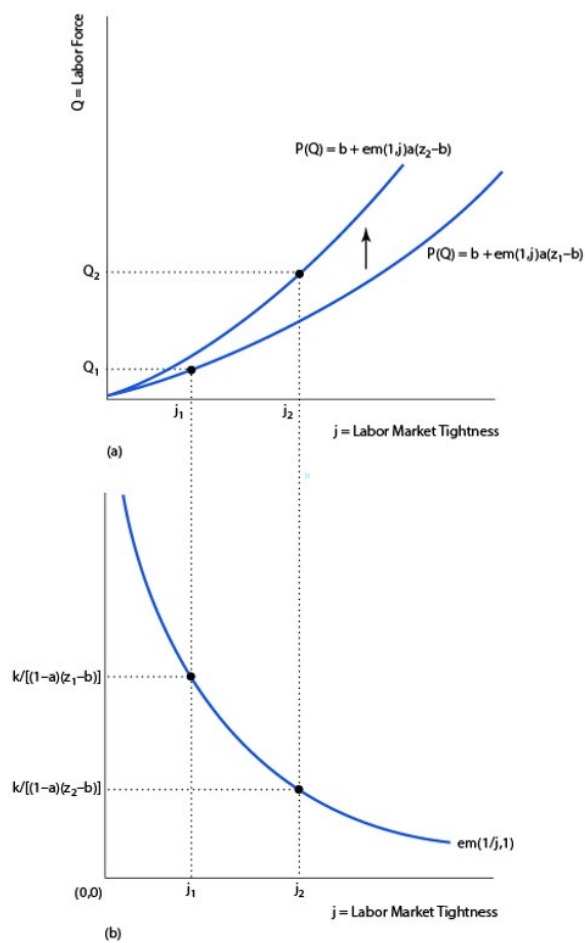
Figure 6.21 An Increase in the UI Benefit (b)



An Increase in Productivity

- Increases the total surplus from a match, $z - b$.
- Increases the wage, w , as the worker gets the same share of a larger pie.
- As profit is higher, posting vacancies becomes more attractive for firms, so labor market tightness, j , rises.
- For consumers, searching for work becomes more attractive, as the wage is higher, and the chances of finding work are better.
- Q rises, u falls, v rises, Y rises.

Figure 6.22 An Increase in Productivity



A Decrease in Matching Efficiency (1 of 2)

- No change in total surplus, or in the wage.
- Chances of finding a worker are lower, so fewer firms post vacancies and j falls.
- For consumers searching is less attractive – the wage is the same, but the chances of finding a job are lower, so Q falls.
- u rises, but vacancy rate stays the same, and Y falls.
- Potential explanation for the shifting Beveridge curve.

A Decrease in Matching Efficiency (2 of 2)

